In [29]:
```
from factor_diffusion_train import FACTOR_NAMES, PREFIX, MODE
from factor_model import FactorModel, load_model
from factor_diffusion_sample import NUM_GENERATE
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
```

In [30]:
```
model = load_model(PREFIX)
print(PREFIX)
```

```
model/regression
```

In [31]:
```
test = pd.read_csv(f"{PREFIX}/test_factors.csv")[FACTOR_NAMES].values
```

Factor Level Evaluation

In [32]:
```
sampled = np.load(f"{PREFIX}/samples/factor_{MODE}_{NUM_GENERATE}.npy")
print(f"{PREFIX}/samples/factor_{MODE}_{NUM_GENERATE}.npy")
```

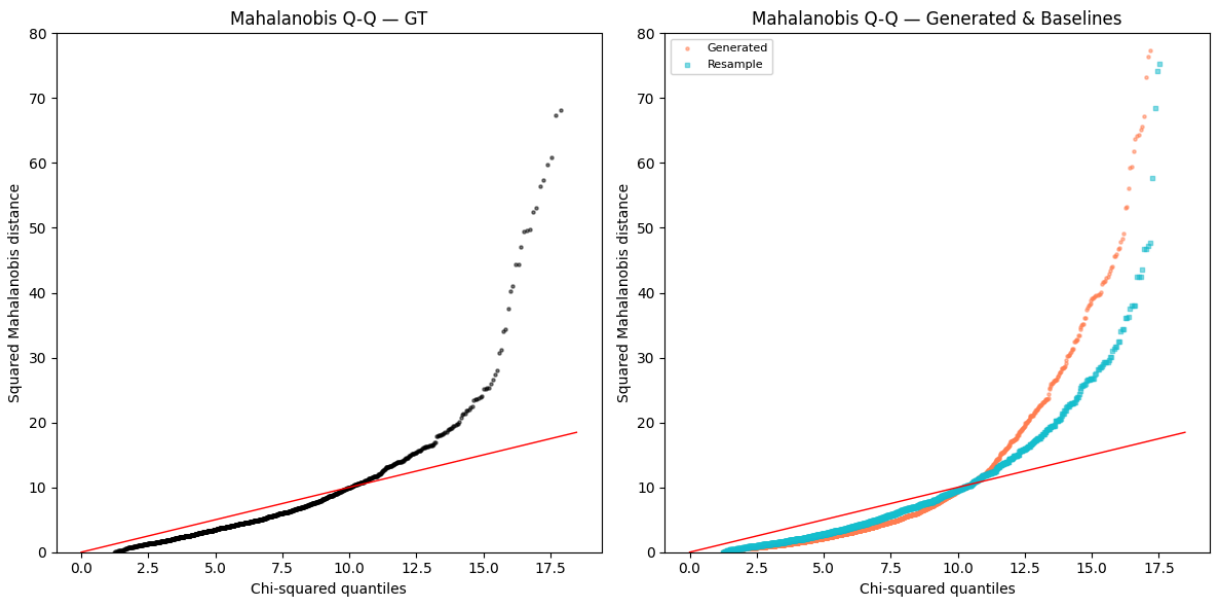
```
model/regression/samples/factor_DLPM_2048.npy
```

In [33]:
```
train = model.F
train = train.drop(columns=['alpha']).values
GT = test

N_BASELINES = 1
rng = np.random.default_rng(42)
baselines = []

row_idx = rng.choice(train.shape[0], size=len(sampled), replace=True)
label = "Resample"
baselines.append((label, train[row_idx]))

base_colors = ['#17becf']

print(f"GT: {GT.shape}  sampled: {sampled.shape}  train: {train.shape}")
print(f"baselines: {N_BASELINES} × {baselines[0][1].shape}")
```

```
GT: (1174, 7)  sampled: (2048, 7)  train: (2738, 7)
baselines: 1 × (2048, 7)
```

In [34]:
```
# 2.0 Mahalanobis check
def mahal_d2(X):
    mean = X.mean(axis=0)
    cov  = np.cov(X, rowvar=False)
    diff = X - mean
    return (diff @ np.linalg.inv(cov) * diff).sum(axis=1)

def plot_mahal(ax, X, label, color='steelblue', marker='o', size=5, alpha=0.
    p  = X.shape[1]
    d2 = mahal_d2(X)
    qs = stats.chi2.ppf(np.linspace(0.01, 0.99, len(d2)), df=p)
    ax.scatter(qs, np.sort(d2), s=size, alpha=alpha, color=color,
```

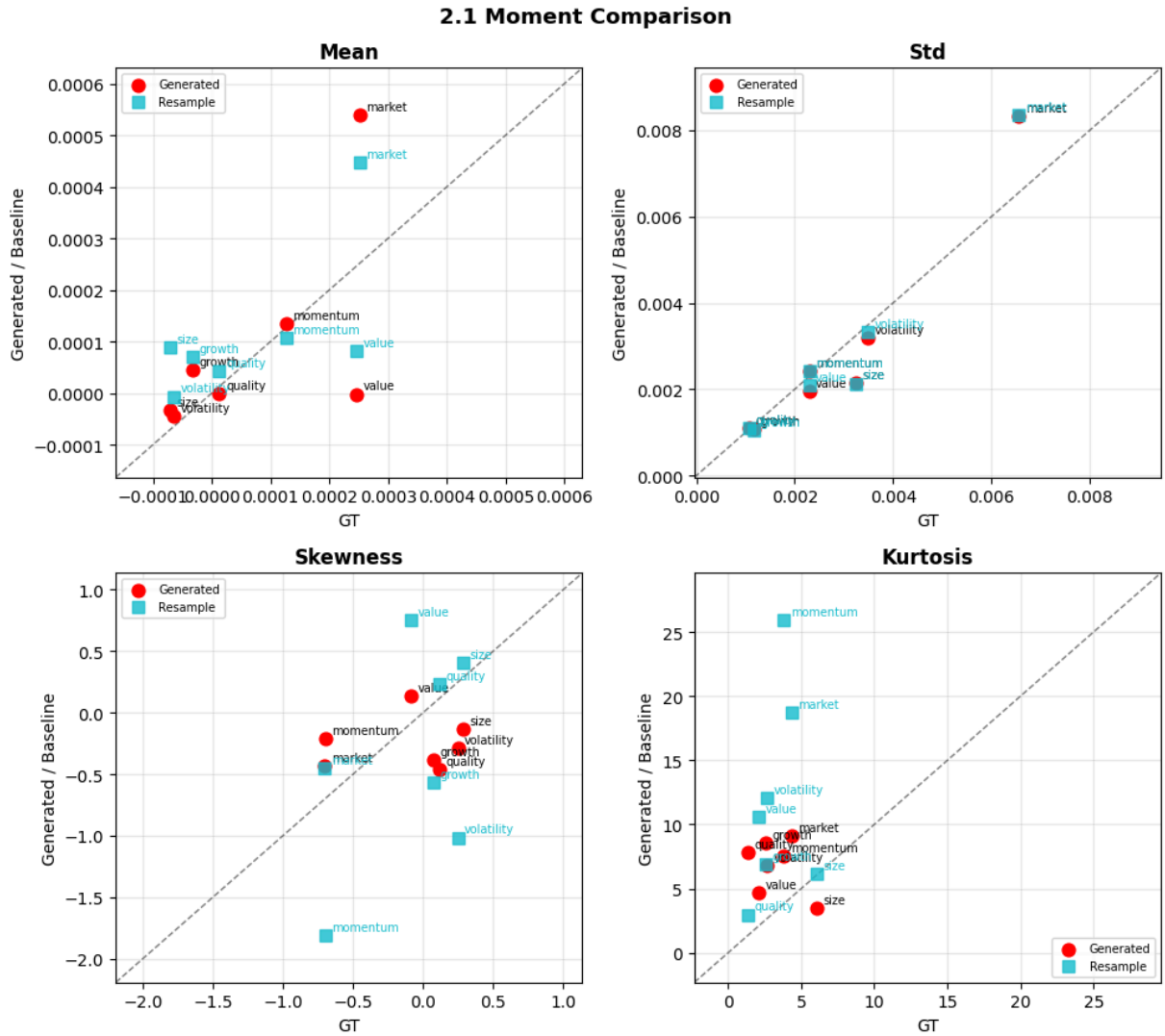
```
                marker=marker, label=label)
    return qs.max()

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 6))

# Left panel: GT
q_max = plot_mahal(ax1, GT, 'GT', color='black')
ax1.plot([0, q_max], [0, q_max], color='red', linewidth=1)
ax1.set_ylim(0, 80)
ax1.set_xlabel('Chi-squared quantiles'); ax1.set_ylabel('Squared Mahalanobis
ax1.set_title('Mahalanobis Q-Q — GT')

# Right panel: Generated + Baselines overlaid
q_max2 = plot_mahal(ax2, sampled, 'Generated', color='coral')
for (bname, bdata), bc in zip(baselines, base_colors):
    q_max2 = max(q_max2, plot_mahal(ax2, bdata, bname, color=bc, marker='s')
ax2.plot([0, q_max2], [0, q_max2], color='red', linewidth=1)
ax2.set_ylim(0, 80)
ax2.set_xlabel('Chi-squared quantiles'); ax2.set_ylabel('Squared Mahalanobis
ax2.set_title('Mahalanobis Q-Q — Generated & Baselines')
ax2.legend(fontsize=8)

plt.tight_layout()
plt.show()
```



In [35]:

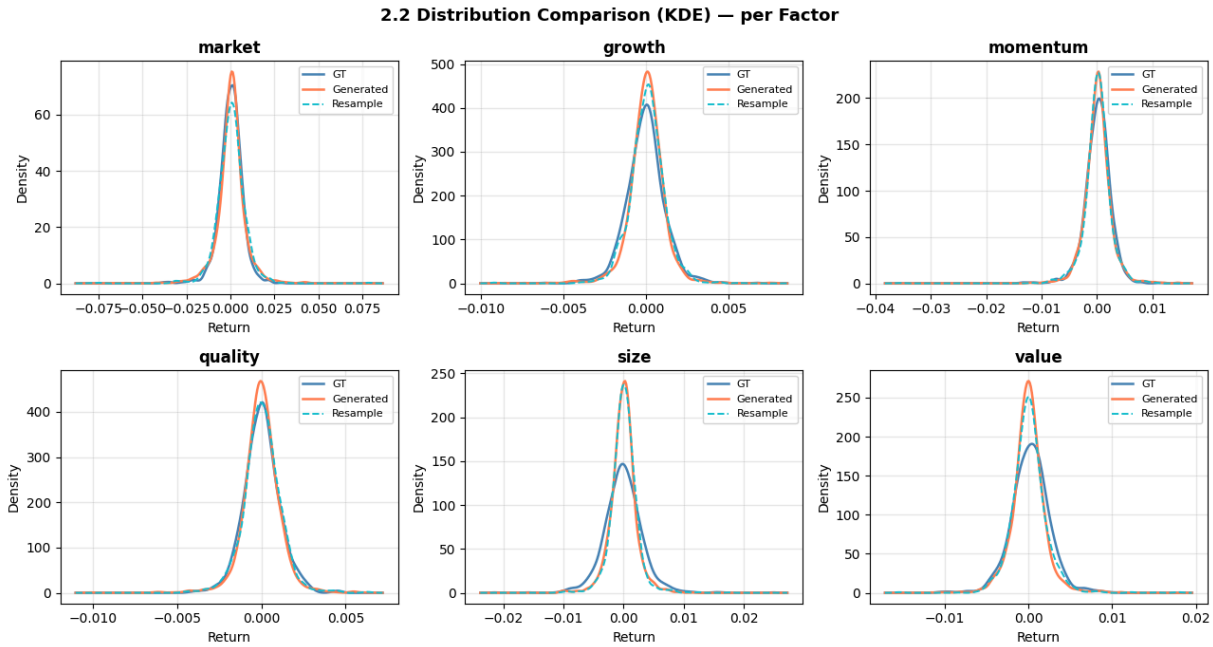
```
# 2.1 Moment Comparison
def compute_moments(data):
    return [np.mean(data, axis=0), np.std(data, axis=0),
            stats.skew(data, axis=0), stats.kurtosis(data, axis=0)]

m_gt    = compute_moments(GT)
m_gen   = compute_moments(sampled)
m_bases = [(name, compute_moments(data)) for name, data in baselines]
moment_names = ['Mean', 'Std', 'Skewness', 'Kurtosis']

fig, axes = plt.subplots(2, 2, figsize=(10, 9))
for idx, (ax, mname) in enumerate(zip(axes.flatten(), moment_names)):
```

```
    gt_m  = m_gt[idx]
    gen_m = m_gen[idx]
    all_y = [gen_m] + [bm[idx] for _, bm in m_bases]

    lim = [min(gt_m.min(), min(y.min() for y in all_y)),
           max(gt_m.max(), max(y.max() for y in all_y))]
    pad = (lim[1] - lim[0]) * 0.15 or 0.01
    lim = [lim[0] - pad, lim[1] + pad]

    ax.plot(lim, lim, 'k--', lw=1, alpha=0.5)

    ax.scatter(gt_m, gen_m, color='red', s=60, zorder=3, label='Generated')
    for i, fn in enumerate(FACTOR_NAMES):
        ax.annotate(fn, (gt_m[i], gen_m[i]), fontsize=7,
                    xytext=(4, 3), textcoords='offset points')

    for (bname, bm), bc in zip(m_bases, base_colors):
        ax.scatter(gt_m, bm[idx], color=bc, s=60, marker='s',
                   zorder=3, label=bname, alpha=0.8)
        for i, fn in enumerate(FACTOR_NAMES):
            ax.annotate(fn, (gt_m[i], bm[idx][i]), fontsize=7,
                        xytext=(4, 3), textcoords='offset points', color=bc)

    ax.set_xlim(lim); ax.set_ylim(lim)
    ax.set_xlabel('GT'); ax.set_ylabel('Generated / Baseline')
    ax.set_title(mname, fontweight='bold')
    ax.grid(True, alpha=0.3)
    ax.legend(fontsize=7)

fig.suptitle('2.1 Moment Comparison', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```



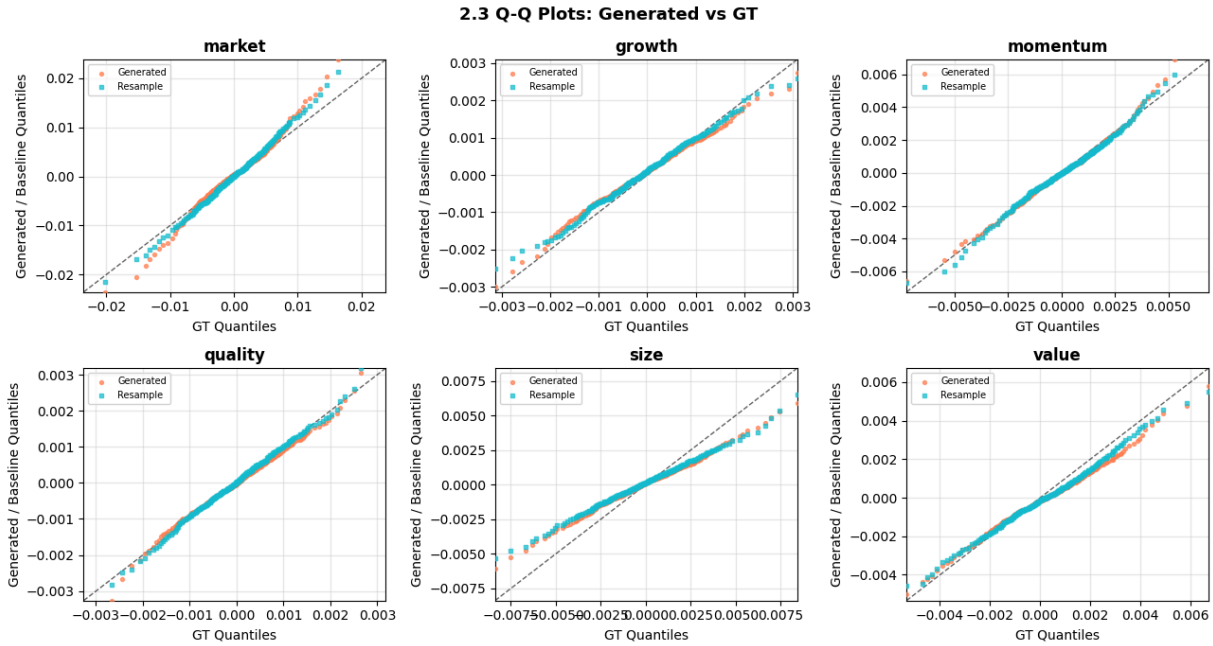
In [36]:
```
# 2.2 Distribution Comparison (KDE)
from scipy.stats import gaussian_kde

n_factors = len(FACTOR_NAMES)
fig, axes = plt.subplots(2, 3, figsize=(13, 7))

for ax, name, i in zip(axes.flatten(), FACTOR_NAMES, range(n_factors)):
    gt_vals  = GT[:, i]
    gen_vals = sampled[:, i]

    all_vals = np.concatenate([gt_vals, gen_vals] +
                              [bdata[:, i] for _, bdata in baselines])
    x_min, x_max = all_vals.min(), all_vals.max()
    pad = (x_max - x_min) * 0.05
    x = np.linspace(x_min - pad, x_max + pad, 300)

    for vals, label, color, ls in [(gt_vals, 'GT', 'steelblue', '-'),
                                   (gen_vals, 'Generated', 'coral', '-')]:
        ax.plot(x, gaussian_kde(vals)(x), label=label, color=color, lw=1.8,

    for (bname, bdata), bc in zip(baselines, base_colors):
        ax.plot(x, gaussian_kde(bdata[:, i])(x),
                label=bname, color=bc, lw=1.4, ls='--')
```

```
    ax.set_title(name, fontweight='bold')
    ax.set_xlabel('Return'); ax.set_ylabel('Density')
    ax.legend(fontsize=8)
    ax.grid(True, alpha=0.3)

fig.suptitle('2.2 Distribution Comparison (KDE) — per Factor', fontsize=13,
plt.tight_layout()
plt.show()
```



In [37]:
```
# 2.3 Quantile-Quantile (Q-Q) Plots
n_quantiles = 200
probs = np.linspace(0.01, 0.99, n_quantiles)

fig, axes = plt.subplots(2, 3, figsize=(13, 7))

for ax, name, i in zip(axes.flatten(), FACTOR_NAMES, range(len(FACTOR_NAMES)
    q_gt  = np.quantile(GT[:, i],      probs)
    q_gen = np.quantile(sampled[:, i], probs)

    all_q = [q_gt, q_gen] + [np.quantile(bdata[:, i], probs) for _, bdata in
    lim = [min(q.min() for q in all_q), max(q.max() for q in all_q)]

    ax.scatter(q_gt, q_gen, s=8, color='coral', zorder=3, label='Generated',

    for (bname, bdata), bc in zip(baselines, base_colors):
        q_base = np.quantile(bdata[:, i], probs)
        ax.scatter(q_gt, q_base, s=8, color=bc, zorder=3,
                   label=bname, alpha=0.7, marker='s')

    ax.plot(lim, lim, 'k--', lw=1, alpha=0.6)
    ax.set_xlim(lim); ax.set_ylim(lim)
    ax.set_title(name, fontweight='bold')
    ax.set_xlabel('GT Quantiles')
    ax.set_ylabel('Generated / Baseline Quantiles')
    ax.legend(fontsize=7)
```

```
    ax.grid(True, alpha=0.3)

fig.suptitle('2.3 Q-Q Plots: Generated vs GT', fontsize=13, fontweight='bolc
plt.tight_layout()
plt.show()
```



In [38]:

```
# 2.4 Kolmogorov-Smirnov Test
from scipy.stats import ks_2samp

D_THRESHOLD = 0.05
n_eff = len(GT) * len(sampled) / (len(GT) + len(sampled))

ks_gen   = [ks_2samp(GT[:, i], sampled[:, i])[0] for i in range(len(FACTOR_N
ks_bases = [(bname, [ks_2samp(GT[:, i], bdata[:, i])[0]
                     for i in range(len(FACTOR_NAMES))])
            for bname, bdata in baselines]

def fmt(stat): return f"{stat:.4f} {'pass' if stat < D_THRESHOLD else 'FAIL'

col_w = 16
header = f"{'Factor':<18} {'Generated':<{col_w}}" + \
         "".join(f"{bname:<{col_w}}" for bname, _ in ks_bases)
print(f"KS Stat (threshold D<{D_THRESHOLD})   n_eff={n_eff:.0f}")
print(header)
print("-" * len(header))
for j, name in enumerate(FACTOR_NAMES):
    row = f"{name:<18} {fmt(ks_gen[j]):<{col_w}}"
    for _, bks in ks_bases:
        row += f"{fmt(bks[j]):<{col_w}}"
    print(row)
```

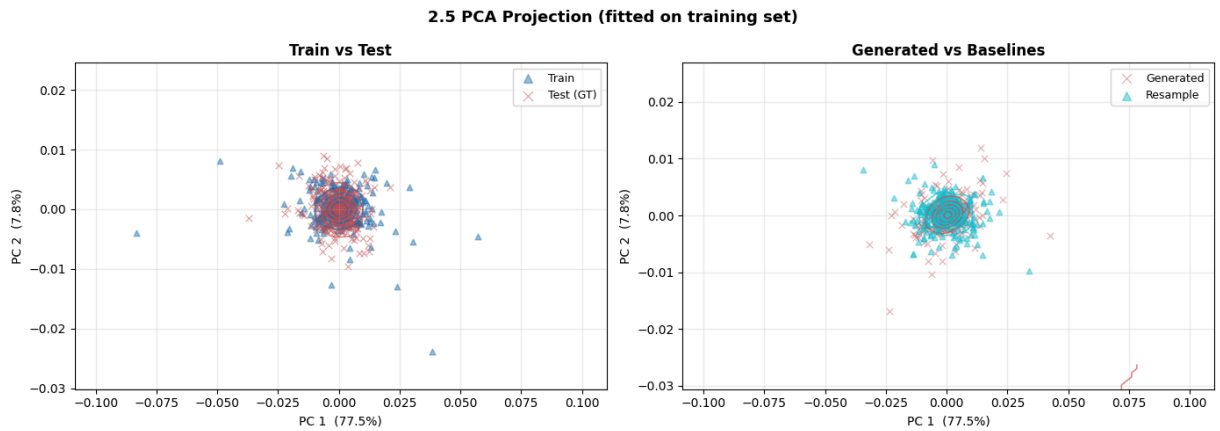
```
KS Stat (threshold D<0.05)   n_eff=746
Factor              Generated       Resample
---------------------------------------------------
market              0.0388 pass     0.0555 FAIL
growth              0.0753 FAIL     0.0692 FAIL
momentum            0.0567 FAIL     0.0465 pass
quality             0.0356 pass     0.0315 pass
size                0.1108 FAIL     0.1306 FAIL
value               0.1189 FAIL     0.1013 FAIL
volatility          0.0832 FAIL     0.0733 FAIL
```

Dimensionality Reduction

In [39]:

```
# 2.5 PCA Projection
from sklearn.decomposition import PCA
from scipy.stats import gaussian_kde

pca = PCA(n_components=2)
pca.fit(train)

train_pc = pca.transform(train)
gt_pc    = pca.transform(GT)
gen_pc   = pca.transform(sampled)
base_pcs = [(bname, pca.transform(bdata)) for bname, bdata in baselines]

xlab = f"PC 1  ({pca.explained_variance_ratio_[0]*100:.1f}%)"
ylab = f"PC 2  ({pca.explained_variance_ratio_[1]*100:.1f}%)"

N_SUB   = 300
C1, C2  = '#1a6faf', '#c44e52'
MARKERS = ['^', 'P', 'D', 's']

def _sub(arr, n=N_SUB, seed=0):
    idx = np.random.default_rng(seed).choice(len(arr), size=min(n, len(arr))
    return arr[idx]

def _contour(ax, pc, color, levels=5, lw=1.2):
    kde  = gaussian_kde(pc.T)
    pad  = (pc.max(axis=0) - pc.min(axis=0)) * 0.15
    lo, hi = pc.min(axis=0) - pad, pc.max(axis=0) + pad
    xx, yy = np.mgrid[lo[0]:hi[0]:80j, lo[1]:hi[1]:80j]
    z = kde(np.vstack([xx.ravel(), yy.ravel()])).reshape(xx.shape)
    ax.contour(xx, yy, z, levels=levels, colors=[color], alpha=0.65, linewic

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# ── Plot 1: Train (blue △) vs Test (crimson ×) ──────────────────────────
_contour(ax1, train_pc, C1)
_contour(ax1, gt_pc,    C2)
ax1.scatter(*_sub(train_pc, seed=0).T, c=C1, marker='^', s=22, alpha=0.45, l
ax1.scatter(*_sub(gt_pc,    seed=1).T, c=C2, marker='x', s=28, alpha=0.55,
            label='Test (GT)', linewidths=0.8)
ax1.set_xlabel(xlab); ax1.set_ylabel(ylab)
ax1.set_title('Train vs Test', fontweight='bold')
ax1.legend(fontsize=9, markerscale=1.5)
```

```
ax1.grid(True, alpha=0.25)

# ── Plot 2: Generated (crimson ×) vs Baselines (blue △ …) ──────────────
_contour(ax2, gen_pc, C2)
ax2.scatter(*_sub(gen_pc, seed=0).T, c=C2, marker='x', s=28, alpha=0.45,
            label='Generated', linewidths=0.8)
for i, ((bname, bpc), bc) in enumerate(zip(base_pcs, base_colors)):
    _contour(ax2, bpc, bc)
    ax2.scatter(*_sub(bpc, seed=i+1).T, c=bc, marker=MARKERS[i % len(MARKERS
                s=22, alpha=0.45, label=bname)
ax2.set_xlabel(xlab); ax2.set_ylabel(ylab)
ax2.set_title('Generated vs Baselines', fontweight='bold')
ax2.legend(fontsize=9, markerscale=1.5)
ax2.grid(True, alpha=0.25)

fig.suptitle('2.5 PCA Projection (fitted on training set)', fontsize=13, for
plt.tight_layout()
plt.show()
```



In [40]:
```
jupyter nbconvert --to webpdf evaluation.ipynb
```

```
  Cell In[40], line 1
    jupyter nbconvert --to webpdf evaluation.ipynb
             ^
SyntaxError: invalid syntax
```